

Gareth Beall

Lead AI/ML Engineer | Applied AI and production systems

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I conceived and built IRIS to help clinicians find reliable answers in scattered hospital documents. The service is in clinical use following a rollout covering 5,000 Queensland Health clinicians. After the first month of wider use, 98% of surveyed clinicians would recommend it.

Experience

Lead AI/ML Engineer | Sunshine Coast Hospital and Health Service

February 2025 to present | Queensland Health

Primary engineer for a hospital AI programme with multimillion-dollar funding. Defined requirements with cybersecurity, hospital and statewide document custodians, and clinical departments.

- In a comparison involving 200 clinicians answering set clinical questions, time to a correct answer averaged about 27 minutes with the existing hospital search portal; with IRIS, most answers were found in under one minute.

Document processing, retrieval and context

- Unified over 10,000 guidelines across PDFs, Word files and web pages. Built collection, scraping and update workflows, reconciling poor structure while retaining source identity and history. Combined vision-language and language models to use image and text evidence.
- Built hybrid search, reranking and table-row retrieval, with evidence selection, conversation history and token budgets. Linked answers to their supporting sources.
- Worked with clinical departments to handle differing guideline recommendations. Adapted retrieval and answer context to departmental resource preferences, the user, clinical question and patient context.

Application engineering and AI-assisted delivery

- Built TypeScript/Next.js APIs and PostgreSQL/pg-boss processing jobs, with indexing readiness checks, bounded retries, visible failures and recovery.
- Built AI-assisted development gates: linting, type checks, tests, automated PR reviews and CI/CD. Encoded recurring mistakes in regression checks and shared agent skills; added migration and recovery tooling.

Retrieval evaluation and model development

- Deployed matched table evidence in reranker inputs after local evaluation recovered previously missed answers. Kept full source sections available for answer generation and citations.
- Fine-tuned and deployed a reranker, comparing variants on real clinician questions using NDCG@10, MRR, recall and latency. Also used synthetic queries and teacher-labelled training data.
- Checked training/evaluation overlap and grouped related queries. Used paired comparisons, confidence intervals and broad regressions, then clinician-graded and automated answer checks to trace failures and assess improvements.

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IRIS engineering | Queensland Health | Continued

GPU architecture and model serving

- Selected GPUs and designed VM layouts for self-hosted OCR, embeddings, reranking and LLMs, including vLLM. Chose models and external APIs against benchmarks, representative documents, hardware and privacy requirements.
- Operated model services with health checks and recovery. Optimised GPU memory, concurrency, KV-cache use and quantisation; load-tested throughput, tail latency, errors and timeouts.

Clinical workflow, governance and delivery

- Completed the Foundational Artificial Intelligence Risk Assessment (FAIRA) and worked through the proposal with Queensland Health cybersecurity. Deployed a PII-detection model and implemented role-based access controls.

Independent work

Founder | Inky Health

August 2026 to present

- Establishing an independent consultancy focused on document systems and private AI.

Founder and software engineer | E2 Apps

September 2022 to present

- Built Opioid Conversion Calculator to address gaps in existing tools, using my clinical experience to shape its content and workflows. Now with over 50,000 downloads worldwide and climbing.
- Created Namely while my wife was pregnant, designing a baby-name app around features missing from other products and the experience parents needed.
- Built My Restored Photo with AI restoration, uploads, editing, payments and scheduled file deletion.

Earlier independent work | 2019-2021

- Began self-directed AI/ML learning, small projects and hospital workflow automations during this period. This work later led to IRIS.
- Created Find the Treasure to encourage outdoor exploration through competitive mobile treasure hunts. Attracted over 10,000 users and raised more than \$100,000, all for charity.

Technical experience

Software: Python, TypeScript/JavaScript, Dart, Java, Kotlin, SQL; Node.js, React, Next.js, Flutter, SwiftUI; APIs, authentication and integrations.

Cloud and infrastructure: Extensive Azure work; AWS EC2 and GPU instances for language-model training; Firebase, Linux, VM security, vLLM, Docker and Podman.

Data and recovery: PostgreSQL/pgvector, MySQL, BM25, hybrid search, Drizzle and pg-boss; migrations, custom backups, write-ahead logging and Rubrik.

AI engineering: Model selection, RAG, context engineering, OCR, vision-language models, reranker fine-tuning and evaluation; self-hosted services and external APIs; inference optimisation, assessing when quantisation is appropriate, and KV-cache optimisation.

Clinical background and education

Fifteen years in clinical pharmacy, including intensive care.

Bachelor of Pharmacy | The University of Queensland | 2008